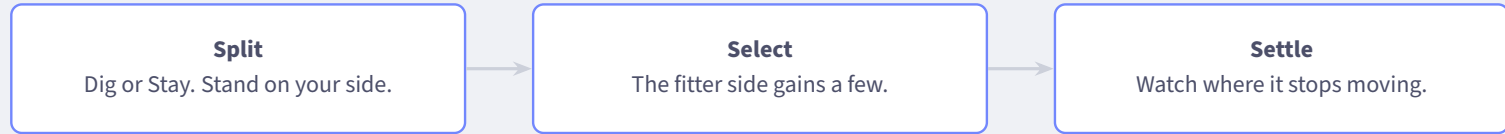


Replicator dynamics

The room is the population



In the room:

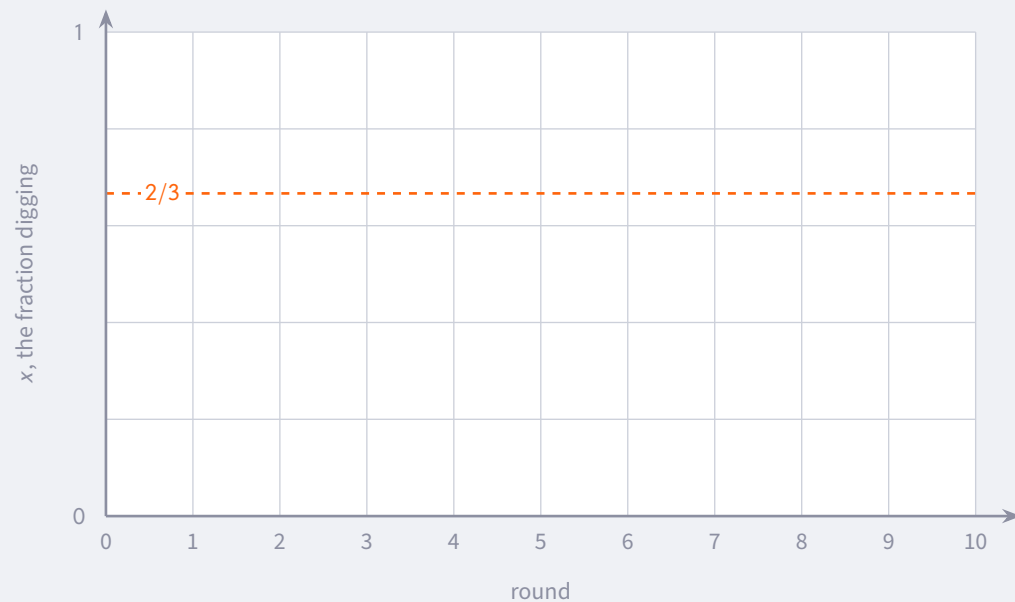
Starting split: digging

	Dig	Stay
Dig	3	2
Stay	4	0

Clearing the drift is worth 4 to each of you. Digging costs 2, shared if you both dig.

Two diggers get 3 each. A lone digger gets while the other free-rides for . Two stayers get .

Would you rather be the one digging?



Plot the split each round. The dashed line is where we expect it to end up.

Round	x	f_D	f_S
1			
2			
3			
4			
5			
6			

With a fraction x digging,

$$f_D = \text{[]}$$

$$f_S = \text{[]}$$

They are equal when $x = \text{[]}$, which is where the room _____.

A strategy grows when it does better than the population average $\phi = x f_D + (1 - x) f_S$. How fast it grows depends on two things:

how far above average it is:

and how many already play it:

$$\dot{x} =$$

Each round of the activity, with its small fixed step, was one step of _____ on this equation.

Factorise $\dot{x}$ and read off every value where it is zero:

Rest points:

For $0 < x < 2/3$ the sign of $\dot{x}$ is , and for $2/3 < x < 1$ it is . So from anywhere inside, the population moves towards

A small group of Stayers invades a population sitting at $x = 2/3$. Do they earn more or less than the residents?

Handwritten notes area.

So $x = 2/3$ _____ an ESS. Rock-Paper-Scissors has an interior rest point too, but there the dynamics _____ instead of settling.

What we just did in the room is **Question 1** on the Replicator Dynamics page.

- (a) The fitnesses f_D and f_S as functions of x . [3]
- (b) The replicator equation, simplified to a product of factors. [7]
- (c) All stable populations. [3]
- (d) From the sign of $\dot{x}$, which one we approach, and where the class settled. [6]
- (e) Whether the interior population is an ESS, argued two ways. [6]

vknigh.org/gt/topics/replicator-dynamics.html